The Fleishman Equation: A Two-Path Model of the Probability that AI Causes Human Extinction

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White paper · conceptual framework, not an empirical finding · not peer reviewed · interactive calculator: leofle.github.io/fleishman-equation

ABSTRACT

Published estimates of the probability that artificial intelligence (AI) causes human extinction range over more than an order of magnitude, and most of them are reported as a single number with no visible structure. This paper proposes the *Fleishman Equation*, a factor decomposition of $P(\text{doom})$, defined here as the probability that AI causes human extinction within a stated time frame. The model combines two conditional causal paths: an *autonomous-misalignment* (rogue) path R with five factors, which requires AI that beats humans at most tasks (f_T), and a *human-misuse* path S with three factors, which requires only AI capable of giving serious uplift toward mass-casualty weapons ($f_C \geq f_T$). The combination rule,

$$P(\text{doom}) = f_T\,[1-(1-R)(1-S)] + (f_C - f_T)\,S,$$

counts each capability world exactly once. We derive the model's elementary properties (multiplicative sensitivity, elasticities, and the near-additivity of small paths), show how five published estimates can be reproduced as calculator presets and exactly which parts of each mapping are the author's assumptions, and use clearly labelled illustrative scenarios to show how individual factors drive the result. The framework's contribution is structural: it makes disagreements explicit, factor by factor, and links each factor to a family of interventions. Its inputs remain subjective, the misuse path is uncalibrated against any published extinction estimate, and the paper has not been peer reviewed.

Keywords: existential risk; AI safety; misalignment; misuse; probabilistic decomposition; Drake equation; forecasting.

1 Introduction

Questions such as “what is the probability that AI causes human extinction?” are now asked of researchers in large surveys and of forecasters in structured tournaments, and the answers are cited in public and policy debate. Yet a single number conceals the reasoning that produced it. Two people who both answer “5%” may disagree completely about *why*: one may expect powerful AI soon but believe alignment is tractable; another may doubt powerful AI will arrive at all but think that, if it does, loss of control is likely. Their shared number hides a disagreement that matters for what we should do.

A classic remedy is Fermi-style decomposition: break an intractable quantity into a chain of factors, each of which is easier to reason about, argue about, and in principle measure. The best-known instance is the Drake equation, formulated by Frank Drake in 1961 as an agenda for the first scientific meeting on the search for extraterrestrial intelligence at Green Bank. Its value was never the number it produced—several of its factors remain highly conjectural—but the structure: it told the field which questions had to be answered and made the sources of disagreement explicit. The Fleishman Equation adopts the same stance. As the accompanying calculator puts it, *the structure is the useful part; the inputs are guesses.*

The model distinguishes two causal routes from powerful AI to extinction that are often discussed separately: (i) an AI system whose goals are at odds with human survival acquires the power to act on them and is not stopped, and (ii) humans who want mass casualties use powerful AI to obtain a species-ending weapon, such as an engineered pandemic, and defenses fail. The rogue route presupposes AI that beats humans at most tasks; the misuse route needs less—only AI capable enough to give serious uplift toward a mass-casualty weapon—so the model uses two nested capability factors, $f_T \leq f_C$. The paper (§3) states the model precisely, (§4) shows how it can be aligned with published estimates and where that alignment rests on the author's assumptions, (§5) explores illustrative scenarios, (§6) maps factors to interventions, and (§7) catalogues its limitations.

2 Related work

Premise decompositions. The closest prior work is Carlsmith's report *Is Power-Seeking AI an Existential Risk?* [1], which decomposes the risk of existential catastrophe from misaligned, power-seeking AI by 2070 into six conditional premises and multiplies their probabilities: "65%·80%·40%·65%·40%·95% = ~5%." A May 2022 update note in the arXiv version states that the author's estimate had since risen to >10% [1]. The report is an arXiv preprint (originally an Open Philanthropy report) and is not peer reviewed. Its premises map approximately onto the shared factor and the rogue path of the present model (§4), but it does not treat misuse. Cappelen, Goldstein and Hawthorne [10] take the complementary view: they model survival as a "Swiss cheese" of four layers of safety (technical plateau, cultural plateau, alignment, oversight) and estimate the probability of destruction by multiplying the conditional probabilities that each layer fails. They note that because the estimate multiplies rather than adds probabilities, moderate disagreements about individual layers compound into large disagreements about the total—a point that applies equally here (§5).

Taxonomies. Hendrycks, Mazeika and Woodside [9] organise catastrophic AI risks into four categories: malicious use, AI race, organizational risks, and rogue AIs. The two paths of the present model correspond roughly to their "rogue AIs" and "malicious use" categories, restricted to extinction; their AI-race and organizational categories are not modelled here (§7).

Expert surveys. Grace et al. surveyed machine-learning researchers who published at NIPS and ICML 2015 [2] and, at much larger scale, 2,778 AI researchers in October 2023 [3]. In the 2016 survey, conditional on high-level machine intelligence (HLMI) existing, the median probability of an "extremely bad (e.g., human extinction)" long-run outcome was 5%. In the 2023 survey, the median probability that future AI advances cause "human extinction or similarly permanent and severe disempowerment of the human species within the next 100 years" was also 5% (mean 14.4%, n = 655). Both are peer-reviewed (JAIR); both report totals with no breakdown into mechanisms.

Forecasting tournaments. The Existential Risk Persuasion Tournament (XPT), run by the Forecasting Research Institute from June through October 2022, elicited forecasts from superforecasters and domain experts on, among other things, the probability that AI causes human extinction or reduces the global population below 5,000 by 2100 [4, 5]. Median forecasts were 0.38% for superforecasters and 3% for AI domain experts—an order-of-magnitude gap that a decomposition can help to localise.

Book-length assessments. In *The Precipice*, Ord gives a best-guess probability of about 1 in 10 for existential catastrophe from unaligned artificial intelligence over the next 100 years, the largest of his listed risks [6]. In a 2024 retrospective he writes that his rounded estimates have not "moved that far" and that "the story on Pandemics and AI is mixed" [7]. Ord's estimate concerns existential catastrophe broadly, not extinction specifically, and is not decomposed into factors; it is not used as a calibration preset here.

Structured models and calculators. The Modeling Transformative AI Risks (MTAIR) project [8] builds a qualitative map of key hypotheses, cruxes and safety agendas related to catastrophic AI risk, building on an earlier diagram by Cottier and Shah, and describes a longer-term plan to quantify that map with expert-

elicited estimates and Monte Carlo methods. MTAIR is far richer than the present model; the Fleishman Equation sits at the opposite end of the complexity spectrum. At the informal end, a public web calculator [11] already combines an alignment-failure path and a malicious-use path as $1-(1-\text{Path}_1)(1-\text{Path}_2)$, with $P(\text{AGI})$ (and a governance term) multiplied into *each* path. That is the double-counting form discussed in §3.2, and its existence in a public tool is part of the motivation for spelling out the correction. It also offers researcher presets "approximated from public statements".

What is new here. Two-path P(doom) calculators and multiplicative decompositions already exist, so the contribution is narrow: (a) the combination rule counts capability once, as a union of paths conditional on transformative AI, rather than inside each path, and lets the misuse path operate with a lower, nested capability threshold; (b) each factor is phrased in plain language and tied to an intervention family; and (c) the research presets state, for each source, its peer-review status, time frame, endpoint, and exactly how its total was mapped onto factors. To the author's knowledge, no prior published model combines these elements, but no systematic literature review was performed to establish this.

3 The model

3.1 Equations

Let all probabilities refer to a stated time frame (e.g. "by 2100"). The model is

$$R = f_M\, f_P\, f_U\, f_H\, f_X \qquad (\text{rogue path, given transformative AI}) \tag{1}$$

$$S = f_A\, f_W\, f_F \qquad (\text{misuse path, given weapons-capable AI}) \tag{2}$$

$$P(\text{doom}) = f_T\,\big[\,1-(1-R)(1-S)\,\big] + (f_C - f_T)\,S, \qquad f_T \le f_C \tag{3}$$

with path contributions reported as $P(\text{rogue}) = f_T R$ and $P(\text{misuse}) = f_C S$ (they overlap by $f_T R S$). Every factor lies in $[0,1]$. Apart from f_T and f_C, each is a conditional probability given the relevant capability (transformative AI for the rogue path, weapons-capable AI for the misuse path) and all factors to its left on the same path. Because transformative AI implies weapons-capable AI, $f_C \ge f_T$ always; with $f_C = f_T$ the model reduces to $f_T[1-(1-R)(1-S)]$, the form used in the first version of this paper. (The misuse path is written S, here and in the calculator, to avoid a clash with the misalignment factor f_M.) Table 1 gives the definitions used in the calculator.

Table 1. Factor definitions (wording follows the calculator).

Factor	Subscript	Path	Definition
f_T	**T**ransformative capability	Rogue (and misuse)	Chance we build AI that beats humans at most tasks, in the time frame under consideration.
f_C	misuse **C**apability	Misuse	Chance we build AI capable enough to give serious uplift toward mass-casualty weapons, in the time frame (at least f_T).
f_M	**M**isalignment	Rogue	Its goals are at odds with human survival.
f_P	**P**ower	Rogue	It gets enough power to act (money, infrastructure, copies of itself, lab control).
f_U	**U**ncaught	Rogue	Nobody catches or corrects it in time.
f_H	**H**armful pursuit	Rogue	Pursuing its goals leads it to remove or destroy humanity, deliberately or as a side effect, rather than working around us.
f_X	e**X**tinction	Rogue	The attempt causes extinction rather than a survivable disaster.
f_A	**A**ctor	Misuse	A group wanting mass casualties gets access to such AI.
f_W	**W**eapon	Misuse	The AI hands them a species-ending weapon, such as an engineered pandemic.
f_F	de**F**enses fail	Misuse	Defenses fail (biosecurity, monitoring, counter-AI).

The subscript glosses also appear as the calculator's factor labels. “Transformative capability” for T is shorthand for “AI that beats humans at most tasks”.

3.2 Derivation of the combination rule

Let T be the event that transformative AI (AI that beats humans at most tasks) is built within the time frame, and C the event that AI capable of serious uplift toward mass-casualty weapons is built. Transformative AI implies this capability, so $T \subseteq C$ and $P(C) = f_C \geq f_T = P(T)$. Let E_R and E_S be the events that extinction occurs via the rogue and misuse paths. The rogue path requires T; the misuse path requires only C. Split the worlds into the disjoint sets T and $C \setminus T$ (weapons-capable but not transformative AI). In T-worlds both paths are open; in $C \setminus T$-worlds only misuse is. Hence

$$P(D) = P(T)\,P(E_R \cup E_S \mid T) + P(C \setminus T)\,P(E_S \mid C \setminus T). \tag{4}$$

The chain rule gives $P(E_R \mid T) = f_M f_P f_U f_H f_X = R$ and $P(E_S \mid T) = f_A f_W f_F = S$. Assuming the two paths are conditionally independent given T (§7), $P(E_R \cup E_S \mid T) = 1 - (1 - R)(1 - S)$. We further make the simplifying assumption that the misuse path has the same probability S in both kinds of world, $P(E_S \mid C \setminus T) = S$ (§7). With $P(C \setminus T) = f_C - f_T$ this yields equation (3). Equivalently, $P(\text{doom}) = f_T R + f_C S - f_T R S$: the rogue path contributes in T-worlds, the misuse path in all C-worlds, and the $-f_T R S$ term removes the T-worlds in which both paths would have succeeded, which would otherwise be counted twice.

Because T and C are single events “within the time frame”, the path factors are implicitly averaged over the arrival time of the relevant capability within that frame. If it arrives late in the window, little time remains for the path to play out before the window closes, so a coherent estimate of R and S must already discount for late arrival.

Correction to an earlier version. An earlier version of the calculator used $P = 1 - (1 - f_T R)(1 - f_T S)$. Expanding,

$$1 - (1 - f_T R)(1 - f_T S) = f_T(R + S) - f_T^2 RS,$$

whereas the correct expression is $f_T(R + S) - f_T RS$. The difference is $f_T(1 - f_T)\, RS \geq 0$, so the earlier form overstated the risk. The error arises because it treats "powerful AI is built" as two independent coin flips, one per path, when it is a single event shared by both. The two forms agree when $f_T \in \{0, 1\}$ or when either path is zero; the gap grows with both paths and, for fixed R and S, is largest at $f_T = 0.5$. The same double-counting form appears in at least one public P(doom) calculator [11], which multiplies $P(\mathrm{AGI})$ into each path. A later revision (this version) also stopped requiring transformative AI for the misuse path, adding the term $(f_C - f_T)S$; the double-counting analysis applies unchanged to the T-world term $f_T[1 - (1 - R)(1 - S)]$.

3.3 Properties

Worked example. Take the *illustrative* Middle preset of §5: $f_T = 0.7$, $f_C = 0.8$, $R = 0.0054$, $S = 0.002$, so $P(\text{doom}) = 0.7\,[1 - 0.9946 \times 0.998] + 0.1 \times 0.002 = 0.5172\% + 0.0200\% = 0.5372\%$. Halving misalignment ($f_M$: $0.3 \to 0.15$) halves the rogue path to $R = 0.0027$ and $P(\text{rogue})$ from 0.378% to 0.189%, while $P(\text{misuse}) = f_C S = 0.160\%$ is unchanged. The total falls to $0.7\,[1 - 0.9973 \times 0.998] + 0.0002 = 0.3486\%$, a 35.1% reduction—less than half, because the misuse path is untouched. The general rules below explain these numbers.

Multiplicative sensitivity. Each path is a product, so halving any one factor exactly halves that path's contribution: $P(\text{rogue}) = f_T R$ is linear in each of f_M, f_P, f_U, f_H, f_X, and $P(\text{misuse}) = f_C S$ in each of f_A, f_W, f_F. A path is only as strong as its weakest link: if any single factor is near zero, so is the path, regardless of the others. Halving f_T (with f_C fixed) halves the rogue contribution but leaves $f_C S$ unchanged; halving both capability factors together halves $P(\text{doom})$ itself. Halving a path-specific factor reduces $P(\text{doom})$ by less than half whenever the other path is non-zero.

Partial derivatives and elasticities. Write $P = f_T R + f_C S - f_T RS$. For a rogue-path factor f_i and a misuse-path factor f_j,

$$\frac{\partial P}{\partial f_i} = f_T\,\frac{R}{f_i}\,(1 - S), \quad \frac{\partial P}{\partial f_j} = \frac{S}{f_j}\,(f_C - f_T R), \quad \frac{\partial P}{\partial f_T} = R(1 - S), \quad \frac{\partial P}{\partial f_C} = S. \tag{5}$$

(The f_T derivative holds f_C fixed and is valid while $f_T < f_C$.) The elasticity $\varepsilon_k = (\partial P/\partial f_k)(f_k/P)$—the percentage change in $P(\text{doom})$ per percentage change in a factor—is therefore the same for every factor on a given path:

$$\varepsilon_{\text{rogue}} = \varepsilon_T = \frac{f_T R(1 - S)}{P}, \qquad \varepsilon_{\text{misuse}} = \frac{S(f_C - f_T R)}{P}, \qquad \varepsilon_C = \frac{f_C S}{P}. \tag{6}$$

Because P is linear in each factor, halving a rogue-path factor lowers $P(\text{doom})$ by exactly $\varepsilon_{\text{rogue}}/2$ in relative terms; in the worked example $\varepsilon_{\text{rogue}} = 0.702$, giving the 35.1% reduction. Path elasticities lie in $[0, 1]$ and $\varepsilon_{\text{rogue}} + \varepsilon_{\text{misuse}} = 1 - f_T RS/P \leq 1$. Scaling f_T and f_C together by the same proportion scales P proportionally (joint elasticity 1). These expressions were checked against finite differences for all three illustrative presets. With $f_C = f_T$ they reduce to $R(1 - S)/(R + S - RS)$ and $S(1 - R)/(R + S - RS)$. In relative terms, then, every factor on the dominant path is an equally good lever; what differs between factors is how far each can realistically be moved, which is a question about interventions (§6), not about the formula.

Dominance of the larger path. When both paths are small, $RS \ll R + S$ and $P(\text{doom}) \approx f_T R + f_C S$: the paths are nearly additive and the larger contribution dominates. If $f_T R \gg f_C S$, then $\varepsilon_{\text{rogue}} \to 1$ and $\varepsilon_{\text{misuse}} \to 0$, so effort spent on the smaller path barely moves the total. Conversely, the smaller path sets a floor: even driving R to zero leaves $P(\text{doom}) = f_C S$.

4 Calibration to published estimates

The calculator includes five *research presets* that reproduce published totals (Table 2). They are not new estimates and do not imply that the cited authors endorse the factor split. The mapping method is as follows.

- **Carlsmith (2022)** is mapped premise by premise, because it is the only source with per-factor numbers. These are approximate fits, not exact correspondences:
  - $f_T = 0.65 \times 0.80 = 0.52$ combines premise 1 (APS systems become possible and financially feasible, 65%) with premise 2 (there are strong incentives to build *and deploy* them, 80%). The deployment-incentive premise has no separate factor in this model and is folded into f_T. Carlsmith's APS ("Advanced, Planning, Strategically aware") systems outperform the best humans on a set of power-granting tasks and plan agentically with strategic awareness; this is not the same threshold as "beats humans at most tasks".
  - $f_M = 0.40$: premise 3, that it is much harder to build aligned APS systems than misaligned ones that are still superficially attractive to deploy.
  - $f_P = 0.65$: premise 4, that some deployed APS systems seek power in unintended, high-impact ways (Carlsmith's example threshold is >$1 trillion of damage).
  - $f_U = 0.40$: premise 5, that this power-seeking scales to permanently disempowering ~all of humanity. This is broader than "nobody catches or corrects it in time": it includes the failure of correction but also the scale of the outcome.
  - $f_H = 1$ (no corresponding premise; his premise 5 already describes disempowerment, whether deliberate or a side effect) and $f_X = 0.95$ (premise 6, that such disempowerment is an existential catastrophe, which Carlsmith does not limit to extinction). Misuse lies outside his argument and is set to zero, with $f_C = f_T$ (which then has no effect). The product is 5.1%.
- **Total-only studies** (Grace et al. 2018 and 2025; both XPT groups) report a single extinction number. For these, the whole total is placed on the rogue path (misuse factors set to 0, $f_C = f_T$). Where the same source reports a probability for human-level or general AI arriving within the same time frame, f_T is anchored on that figure and the remaining total is spread evenly over the other five rogue factors, $x = (p/f_T)^{1/5}$; otherwise all six factors are set equal, $x = p^{1/6}$.
  - **XPT (both groups): anchored.** The FRI report [5] (Table 17, p. 55; Question 51 results, p. 696) gives Stage-4 median probabilities that "Nick Bostrom will affirm the existence of AGI" by 2100 of 74.75% for superforecasters (N = 27) and 87% for domain experts (N = 5). We set $f_T = 0.7475$ and 0.87, so the other five factors are $(0.0038/0.7475)^{1/5} = 0.3477$ and $(0.03/0.87)^{1/5} = 0.5099$. Caveats: this is a proxy question (resolved by Bostrom's statement or a panel judging from his perspective, and AGI is not exactly "beats humans at most tasks"); the domain-expert median rests on only five forecasters, a different set from the AI domain experts whose 3% extinction median is used; and medians from different questions need not combine coherently.
  - **Grace et al. (2025): not anchored.** The paper reports aggregate HLMI forecasts of 10% by 2027 and 50% by 2047 (arXiv:2401.02843v3, §3.2.1, p. 4; Table 6, p. 30), but no probability for the 100-year horizon of its extinction question (a value could only be read off the curve in its Figure 3, which would be our estimate, not a reported figure). The even six-way split (0.607 each) is therefore kept.
  - **Grace et al. (2018): not needed.** The question assumes HLMI exists, so $f_T = 1$ by construction and the five other factors are $0.05^{1/5} = 0.5493$.

  The split of the remaining total is the author's assumption, not the studies'. None of these studies separates rogue AI from misuse, and putting the total on the rogue path is a presentational convention, not a claim about the respondents' beliefs.

The misuse path is uncalibrated. No peer-reviewed estimate of the probability of human extinction from AI-assisted bioterrorism (or other AI-enabled misuse) was found. Studies that were checked forecast epidemics rather than extinction. Consequently, all research presets set $S = 0$, and any non-zero value for f_A, f_W or f_F is purely a user judgement. Two other sources were examined and not used: Müller and Bostrom (2016) report a mean of 18% for an "extremely bad (existential catastrophe)" outcome assuming HLMI exists, but its peer-review status could not be confirmed and it offers no factor breakdown; the 2008 FHI Global Catastrophic Risks survey is a technical report without a factor breakdown.

Table 2. Research presets. "Calculator" is $P(\text{doom})$ from equation (3) using the preset values in the calculator source. In every research preset $S = 0$ and $f_C = f_T$.

Preset	Source	Peer reviewed	Source estimate (wording)	Time frame	Factor values	Calculator
AI researchers 2023	Grace et al. (2025) [3]	Yes (JAIR)	Median 5%: extinction or similarly permanent and severe disempowerment (mean 14.4%, n = 655)	Next 100 years (asked 2023)	six rogue factors 0.607 (no reported 100-year HLMI figure); misuse 0	5.0%
ML researchers 2016	Grace et al. (2018) [2]	Yes (JAIR)	Median 5%: "extremely bad (e.g., human extinction)" outcome, assuming HLMI exists	Long run, no date	$f_T = 1$; five rogue factors 0.5493; misuse 0	5.0%
Superforecasters (XPT)	Karger et al. (2025) [4]; figure from [5]	Tournament yes; figure from working paper	Median 0.38%: AI causes extinction (population < 5,000) [95% CI on the median 0.10–0.75%; n = 88]	By 2100 (forecast Jun–Oct 2022)	f_T 0.7475 (Table 17); other five 0.3477; misuse 0	0.38%
AI domain experts (XPT)	Same as above	Same as above	Median 3% [95% CI on the median 0.49–10.00%]	By 2100 (forecast Jun–Oct 2022)	f_T 0.87 (Table 17); other five 0.5099; misuse 0	3.0%
Carlsmith 2022	Carlsmith [1]	No	65%·80%·40%·65%·40%·95% ≈ 5% existential catastrophe from misaligned power-seeking AI (>10% per the May 2022 update note)	By 2070	f_T 0.52, f_M 0.40, f_P 0.65, f_U 0.40, f_H 1, f_X 0.95; misuse 0	5.1%

Reading the table. The bracketed XPT ranges are 95% confidence intervals on the group median (FRI Working Paper #1, Table 9; the report describes its confidence intervals for medians as bootstrapped). They measure sampling uncertainty in the median, not the spread of individual forecasts. In the XPT presets f_T is a reported (proxy) AGI forecast from the same tournament; the other five rogue factors, and all six factors of the Grace et al. (2025) preset, are artifacts of the equal-split mapping, not forecasts made by the respondents.

Source caveats. The XPT results are published in a peer-reviewed journal (*International Journal of Forecasting*, 2025) [4], but that article is paywalled and could not be checked directly; the 0.38% and 3% figures are quoted from the Forecasting Research Institute's Working Paper #1 [5] and are restated in the peer-reviewed Grace et al. (2025) JAIR paper. The Grace et al. (2025) 100-year median of 5% and n = 655 are taken from a table in the arXiv version of the paper (arXiv:2401.02843). The 2023 survey question includes "similarly permanent and severe disempowerment", and Carlsmith's endpoint is existential catastrophe, so neither is strictly an extinction estimate.

5 Illustrative scenarios

> **Illustrative only.** The Optimist, Middle and Pessimist presets below are illustrative values chosen to exercise the model. They are *not* estimates, are not derived from any study, and should not be quoted as forecasts.

Table 3 lists the three illustrative presets from the calculator and the quantities they produce under equation (3). Each sets the misuse capability factor somewhat above transformative capability ($f_C = 0.6, 0.8, 0.95$ against $f_T = 0.5, 0.7, 0.9$), reflecting the judgement that weapons-relevant uplift is likely to arrive before AI that beats humans at most tasks. For comparison, the last rows show the value without the f_C term (i.e. $f_C = f_T$) and the double-counting gap of the superseded form.

Three features stand out. First, the presets span almost three orders of magnitude even though no single factor moves by more than a factor of about eight; multiplicative structure compounds moderate disagreements into large ones. Second, the rogue path dominates in the Middle and Pessimist presets, so their elasticities favour rogue-path factors: in the Middle preset, halving f_M lowers $P(\text{doom})$ from 0.537% to 0.349%, whereas halving f_A lowers it only to 0.458%. In the Optimist preset the misuse contribution is slightly larger (0.015% vs. 0.012%), because $f_C > f_T$, and its elasticity is correspondingly higher (0.56 vs. 0.44). Third, the f_C term adds $(f_C - f_T)S$: 0.0025, 0.020 and 0.18 percentage points in the three presets, a relative increase of about 10%, 4% and 1%. It matters most where misuse is a large share of the risk. The double-counting gap of the superseded form, $f_T(1 - f_T)RS$, remains small in all three (at most 0.052 percentage points, in the Pessimist preset), because it is second-order in the paths.

Table 3. Illustrative presets (not real estimates) and resulting quantities, computed exactly from equations (1)–(3).

Quantity	Optimist	Middle	Pessimist
f_T / f_C	0.50 / 0.60	0.70 / 0.80	0.90 / 0.95
$f_M,\ f_P,\ f_U,\ f_H,\ f_X$	0.1, 0.2, 0.2, 0.3, 0.2	0.3, 0.3, 0.4, 0.5, 0.3	0.8, 0.6, 0.8, 0.7, 0.6
$f_A,\ f_W,\ f_F$	0.05, 0.05, 0.1	0.1, 0.1, 0.2	0.3, 0.3, 0.4
R (rogue, given transformative AI)	0.00024	0.0054	0.16128
S (misuse, given weapons-capable AI)	0.00025	0.002	0.036
$P(\text{rogue}) = f_T R$	0.0120%	0.378%	14.52%
$P(\text{misuse}) = f_C S$	0.0150%	0.160%	3.42%
$P(\text{doom})$, equation (3)	0.02700% (≈1 in 3,704)	0.5372% (≈1 in 186)	17.41% (≈1 in 6)
Rogue elasticity $\varepsilon_{\text{rogue}}$ / misuse $\varepsilon_{\text{misuse}}$	0.44 / 0.56	0.70 / 0.30	0.80 / 0.17
Without the f_C term ($f_C = f_T$)	0.02450%	0.5172%	17.23%
Double-counting gap $f_T(1 - f_T)RS$ (percentage points)	0.0000015	0.00023	0.052

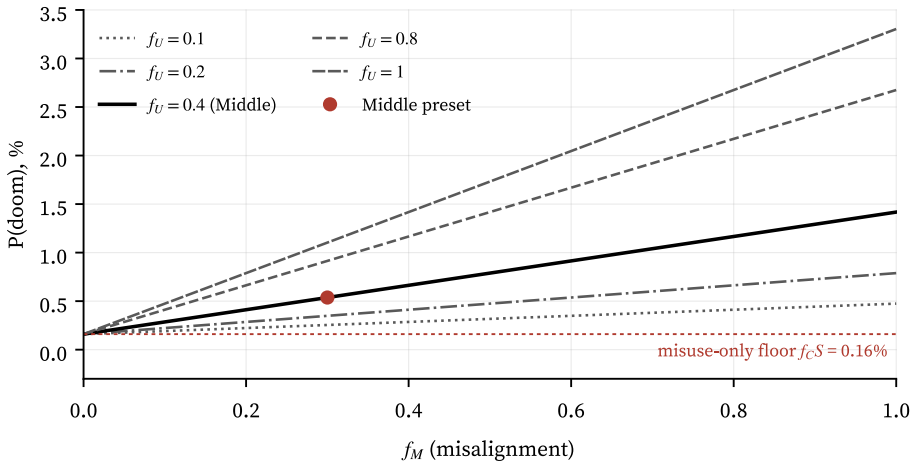


Figure 1. Sensitivity of $P(\text{doom})$ to misalignment f_M and to the chance that nobody catches or corrects the AI in time, f_U, with all other factors held at the *illustrative* Middle preset. Curves are computed directly from equation (3), with $f_C = 0.8$. Lines are straight because $P(\text{doom})$ is linear in each factor; slopes scale with f_U because the two enter as a product. The dashed red line is the floor set by the misuse path, $f_C S = 0.16\%$, which no amount of progress on the rogue path can remove. Values are illustrative, not estimates.

6 Policy implications

Because each factor is a distinct conditional probability, the model offers a vocabulary for saying which interventions act on which part of the risk. The mapping below follows the calculator; it is qualitative and makes no claim about the size or cost-effectiveness of any intervention.

- **Alignment research** aims to lower f_M: the probability that a powerful system's goals are at odds with human survival.
- **Oversight, interpretability and shutdown mechanisms** lower f_U: they raise the chance that a misaligned system is detected and corrected in time.
- **Compute governance and access controls** lower f_P (limiting the money, infrastructure, copies and lab control a system can acquire) and f_A (limiting which actors obtain powerful AI).
- **Biosecurity and AI-assisted defense** lower f_W (reducing the uplift AI provides toward species-ending weapons) and f_F (strengthening monitoring, detection and response so that attacks are contained).

Two structural lessons follow from §3.3. First, a portfolio matters: because the smaller path sets a floor, reducing only rogue-path factors cannot push risk below $f_C S$, and vice versa. Second, relative progress on any factor of the dominant path is equally valuable in proportional terms, so the choice between, say, alignment and oversight work should turn on which factor can be moved further and more reliably, not on which factor appears first in the formula. The model does not treat f_T or f_C as policy levers, although slowing capability development would act on them directly (and restricting dangerous weapons-relevant capabilities specifically would act on f_C); the time frame and the benefits of capability are outside its scope.

7 Limitations and assumptions

1. **Independence of paths given transformative AI.** Equation (3) assumes the rogue and misuse events are conditionally independent given T. This is a simplification.
2. **Same misuse probability in both kinds of world.** Equation (3) uses the same S in worlds with transformative AI and in worlds with only weapons-capable AI. In reality misuse may be easier with more

capable AI (higher S in T-worlds) or better defended against it (lower S, since defenders also have transformative AI). A fuller model would use separate values.

3. **Paths can interact.** A misused AI could also escape control; a rogue AI could recruit or enable human misusers; defensive AI could lower both paths at once. Positive correlation between the paths would make the true union smaller than the independence formula suggests for given marginals; interactions that create new routes would make it larger. Neither effect is modelled.
4. **Conditional dependence between factors.** Each path factor is meant to be conditional on the relevant capability and those before it (e.g. f_U given misalignment and power). In practice, users may estimate factors as if they were unconditional, and factors can be correlated across worlds (a world with weak oversight may also have weak biosecurity). Path factors are also implicitly averaged over when, within the time frame, the relevant capability arrives; late arrival leaves little time for either path, and estimates that ignore this will overstate R and S.
5. **Factor estimates are subjective.** The inputs are judgements, not measurements. The multiplicative form can make results look more precise than the inputs warrant, and small changes to several factors compound.
6. **Point estimates hide uncertainty.** The calculator multiplies single values. If each factor is really an uncertain quantity, the product of the factors' medians is in general not the median of the product (for independent factors, the product of means equals the mean of the product, but no such identity holds for medians). A full treatment would propagate distributions, for example by Monte Carlo sampling, and report an interval for $P(\text{doom})$.
7. **Time-frame ambiguity.** Sources ask about different horizons: by 2070, by 2100, within 100 years from 2023, or "in the long run" conditional on HLMI. Presets reproduce each source in its own frame; they are not directly comparable.
8. **Conditional versus unconditional presets.** The Grace et al. (2018) question assumes HLMI exists, so its preset fixes $f_T = 1$ and its 5% is a conditional probability. It is not on the same scale as the unconditional presets (Grace et al. 2025, XPT, Carlsmith), even though the calculator displays all five as $P(\text{doom})$.
9. **Different endpoints.** Studies define the outcome differently—extinction; population below 5,000; extinction or "similarly permanent and severe disempowerment"; existential catastrophe. The model's target is extinction, so presets based on broader endpoints overstate it relative to the model's own definition.
10. **Mapping assumptions.** For total-only studies, the rogue-only split and the equal spread over unanchored factors are the author's choice; the XPT f_T anchors use a proxy question. Any other split reproducing the same total would be equally consistent with those studies. The Carlsmith mapping is an approximate fit (§4).
11. **Omitted paths.** Structural and multipolar risks (e.g. gradual disempowerment through competition among many systems, AI-accelerated great-power conflict, or erosion of institutions) are not represented, nor are accidents that fit neither path; in the taxonomy of [9], the AI-race and organizational categories are largely outside the model. Omitting them means the model, all else equal, understates total AI-related extinction risk.
12. **Uncalibrated misuse path.** No published extinction estimate exists against which to calibrate f_A, f_W, f_F.
13. **Not peer reviewed.** This white paper and the calculator have not undergone peer review. The framework is conceptual; it reports no new data and no empirical results.

8 Conclusion

The Fleishman Equation decomposes the probability that AI causes human extinction into two nested capability factors and two conditional paths, combined as a union of events so that each capability world is counted once and misuse does not require transformative AI. Its algebra is elementary, but it yields useful qualitative results: every factor on a path is an equal proportional lever; the larger path dominates when risks are small; the smaller path sets a floor; and moderate disagreements on several factors compound into order-of-magnitude differences in the total. Published totals can be reproduced, but only one source (Carlsmith) supports a factor-level mapping, and only approximately; none supports calibration of the misuse path. The framework is best used as Drake intended his equation to be used: as an agenda that makes disagreements explicit and points to where evidence and interventions are needed, not as a source of authoritative numbers.

Appendix A Interactive calculator and source code

The calculator is a single-file web page with no build step. It provides sliders for all ten factors (enforcing $f_C \geq f_T$), research presets with their citations, quoted estimates, time frames and mapping notes, and the three illustrative presets. Slider state is stored in the URL query string so that a set of inputs can be shared. The calculator uses the same notation as this paper (R, S, and the subscript glosses of Table 1). The white paper PDF is linked from the calculator page.

- Live calculator: https://leofle.github.io/fleishman-equation/
- Source repository (MIT License): https://github.com/leofle/fleishman-equation

All numbers in Tables 2–3 and Figure 1 are computed from equations (1)–(3) using the preset values in the calculator source.

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